

Topological-Temporal Learning

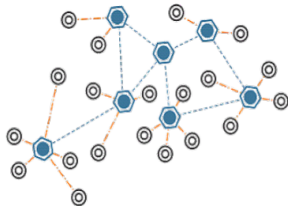
Elvin Isufi

Intelligent Systems Department

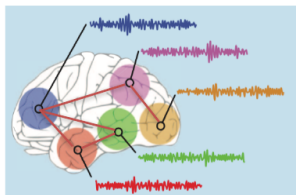
May 15, 2026

The World is Dynamic

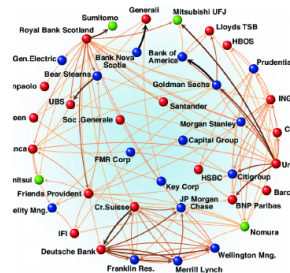
- **Data** on the topological elements are often **time varying**



Monitoring sensors



Brain recordings
Epileptic signals



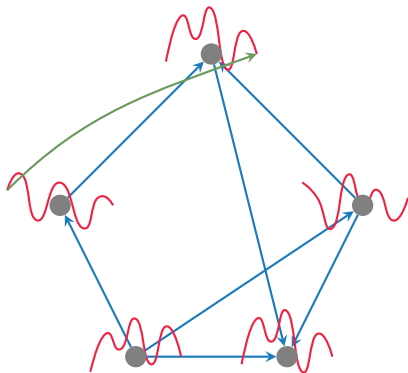
Financial networks

- Data now have a **spatial** (topology) and a **temporal** dependency.

Spatio-Temporal Dependencies

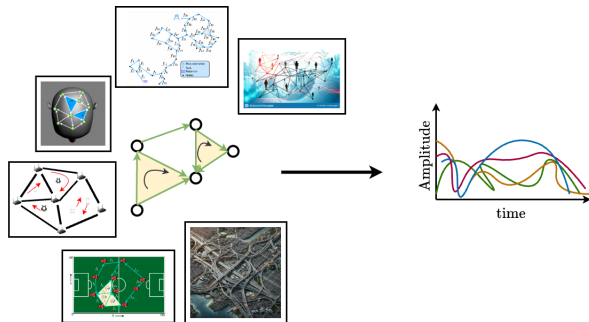
- ▶ **Spatial Dependency:** Future value of node influenced by neighbors at same timestep
- ▶ **Temporal Dependency:** Future value of node influenced by it's own past values
- ▶ **Spatio-Temporal Dependency:** Future value of mode influenced by past values of neighbors

Core Challenge: Effectively integrating spatial, temporal, and joint spatio-temporal interactions



Things Escalate Quickly on Topologies

- **Examples:** Brain network, Water distribution system dynamics, Road transportation network
- **Aim:** Combine the SC structure of network and time-series nature of data for better inference.

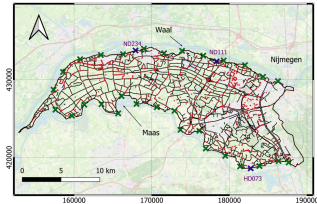


;

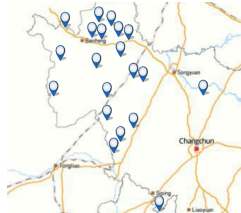
- **Tasks:** Forecasting, spatiotemporal signal inputation, anomaly detection

Machine Learning with Topological Time Series

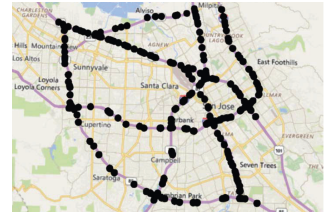
Forecasting in infrastructure network



(a) Water network at Amersfoort, the Netherlands



(b) Locations of wind farms in Jilin, China

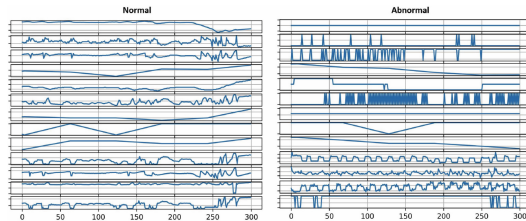
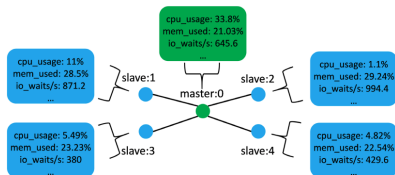


(c) Road sensor network in California, USA.

- ▶ (a) Node-level dike-breach flood prediction using hydraulic structure nodes and edges.
- ▶ (b) Node-level ultra-short-term wind power prediction using historical wind power time-series nodes and spatial edges.
- ▶ (c) Node-level long-term traffic prediction using sensor nodes and road connectivity edges.

Machine Learning with Topological Time Series

Anomaly detection

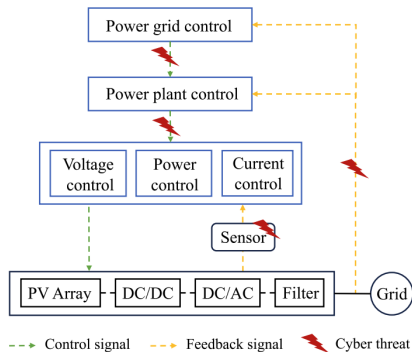


- Anomaly detection in cloud systems
 - ⇒ Master/slave nodes have multivariate time-series system metrics as features
 - ⇒ Topological connectivity and interacting relationships serve as the edges
 - ⇒ Node-level anomaly scores can trigger system-level alert

- Multivariate time-series anomaly detection (engine failures) for predictive vehicle maintenance
 - ⇒ Individual vehicle sensors are the nodes
 - ⇒ The edges represent the statistical dependence between sensors

Machine Learning with Topological Time Series

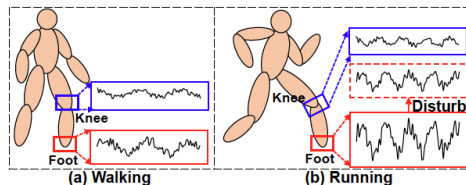
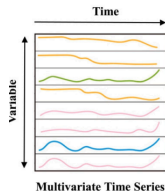
Anomaly detection



- Cyberattack detection in grid-tied photovoltaic (PV) systems
 - ⇒ Signal data from the inter-networked power electronics converters serve as the node features
 - ⇒ Latent spatial correlations and physical system dynamics are used to create a dynamic weighted adjacency matrix
 - ⇒ System-level cyberattacks are detected by extracting spatial correlations and temporal patterns

Machine Learning with Topological Time Series

Time series classification



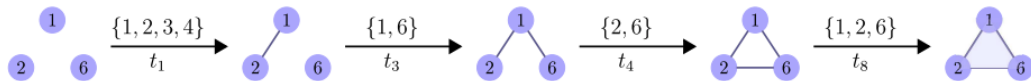
- Multivariate time series classification without predefined graph structures

- Multivariate time series classification with massive unlabeled samples

- Individual sensors (or variables) serve as the nodes
- Correlations and spatial dependencies between the sensors serve as the edges

Machine Learning with Topological Time Series

Higher-order closure prediction

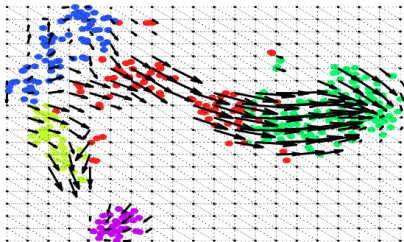


A simplicial closure event for nodes 1, 2, and 6 .

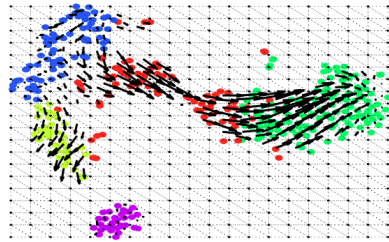
- ▶ Given time series over the nodes only, triangles may form over time;
- ▶ Task: Predict the higher-order topological structure closure
- ▶ Applications:
 - ⇒ Social Networks: Forecasting new triadic follower and friendship links.
 - ⇒ Academia: Predicting multi-author paper collaborations.
 - ⇒ Online Forums: Anticipating which sets of users will jointly participate in a specific thread.
 - ⇒ Entertainment: Predicting collaborations between musical artists.
 - ⇒ etc.

Machine Learning with Topological Time series

Signal interpolation



Partially observed RNA velocity field



Reconstructed RNA velocity field

- ▶ Given coexisting temporal node and edge signals, infer missing values;
- ▶ Applications:
 - ⇒ Transportation Networks: Estimating missing traffic flows on edges
 - ⇒ Epidemiology: Forecasting daily COVID-19 cases (nodes) based on population mobility (edges)

Outline: What is this Lecture About?

- ▶ How to model time series over topologies?

- ⇒ Graph Vector Autoregressive Models

- ⇒ Topological Vector Autoregressive Models

Graph Vector Autoregressive Models

Recursive Models

Autoregressive models for scalar time series - AR(P)



Signal at time t Past signals

↓ ↓

$$x_t = - \sum_{p=1}^P h_p x_{t-p} + \epsilon_t$$

- ▶ $h_1, \dots, h_p$ scalar parameters
- ▶ ϵ_t uncertainty term
- ▶ useful for signal modeling/forecasting

Vector Auto-Regressive (VAR) Model

- ▶ A VAR model expresses a multivariate time-varying process

$$\mathbf{x}_t = \sum_{p=1}^P \mathbf{W}_p \mathbf{x}_{t-p} + \boldsymbol{\varepsilon}_t$$

- ▶ $\mathbf{W}_p = \{w_{mn;p}\}$: weight between time series m and n at instant $t - p$

Vector Auto-Regressive (VAR) Model

- ▶ A VAR model expresses a multivariate time-varying process

$$\mathbf{x}_t = \sum_{p=1}^P \mathbf{W}_p \mathbf{x}_{t-p} + \boldsymbol{\varepsilon}_t$$

- ▶ $\mathbf{W}_p = \{w_{mn;p}\}$: weight between time series m and n at instant $t - p$
- ▶ can be written as $\mathbf{x}_t := \Phi_{\text{VAR}}(\mathbf{x}_{t-1}, \dots, \mathbf{x}_{t-P}; \mathcal{H}_{\text{VAR}})$,
 $\mathcal{H}_{\text{VAR}} = \{\mathbf{W}_1, \dots, \mathbf{W}_P\}$,

Vector Auto-Regressive (VAR) Model

- ▶ A VAR model expresses a multivariate time-varying process

$$\mathbf{x}_t = \sum_{p=1}^P \mathbf{W}_p \mathbf{x}_{t-p} + \boldsymbol{\varepsilon}_t$$

- ▶ $\mathbf{W}_p = \{w_{mn;p}\}$: **weight** between **time series m and n** at instant $t - p$

- ▶ can be written as $\mathbf{x}_t := \Phi_{\text{VAR}}(\mathbf{x}_{t-1}, \dots, \mathbf{x}_{t-P}; \mathcal{H}_{\text{VAR}})$,
 $\mathcal{H}_{\text{VAR}} = \{\mathbf{W}_1, \dots, \mathbf{W}_P\}$,

$$\underset{\mathbf{W}}{\text{minimize}} \sum_{t=P+1}^{|\mathcal{T}|} [l(\mathbf{x}_t, \Phi_{\text{VAR}}(\mathbf{x}_t, \dots, \mathbf{x}_{t-P}, \mathbf{W}))] + \lambda g(\mathcal{H}_{\text{VAR}})$$

Vector Auto-Regressive (VAR) Model

- ▶ A VAR model expresses a multivariate time-varying process

$$\mathbf{x}_t = \sum_{p=1}^P \mathbf{W}_p \mathbf{x}_{t-p} + \boldsymbol{\varepsilon}_t$$

- ▶ $\mathbf{W}_p = \{w_{mn;p}\}$: weight between time series m and n at instant $t - p$
- ▶ can be written as

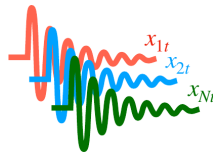
$$\mathbf{x}_t := \Phi_{\text{VAR}}(\mathbf{x}_{t-1}, \dots, \mathbf{x}_{t-P}; \mathcal{H}_{\text{VAR}}),$$

$$\mathcal{H}_{\text{VAR}} = \{\mathbf{W}_1, \dots, \mathbf{W}_P\},$$

$$\underset{\mathbf{W}}{\text{minimize}} \sum_{t=P+1}^{|\mathcal{T}|} [l(\mathbf{x}_t, \Phi_{\text{VAR}}(\mathbf{x}_t, \dots, \mathbf{x}_{t-P}, \mathbf{W}))] + \lambda g(\mathcal{H}_{\text{VAR}})$$

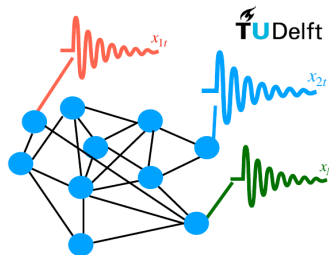
- ▶ Limitations

- ⇒ lack of the structural information
- ⇒ too many parameters ⇒ $\mathbf{W}_p$ are $N \times N$ parameter matrices
- ⇒ lack of structural invariants



VAR for Graph Data

- ▶ When data have a graph structure
 - ⇒ Graph convolutions can be handy
- ▶ How to use them for time series?
 - ⇒ Substitute each $\mathbf{H}_p$ with $\mathbf{H}_p(\mathbf{S}) = \sum_{k=0}^K h_{kp} \mathbf{S}^k$
 - ⇒ Reduce parameters from N^2 to $K + 1$

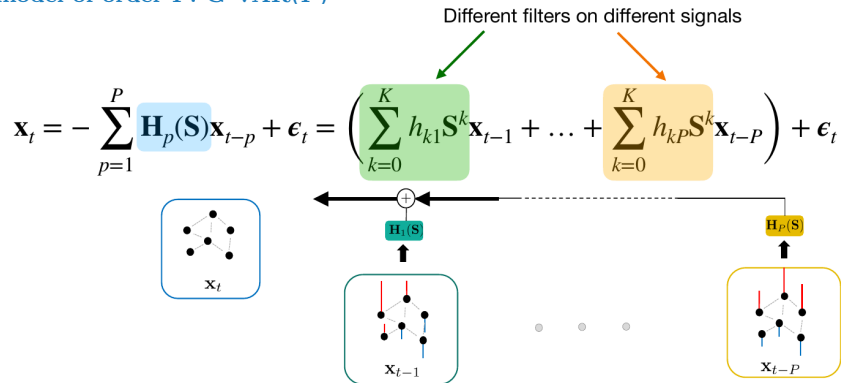


Graph convolutions capture
the structure in data

$$\mathbf{H}(\mathbf{S}) = \sum_{k=0}^K h_k \mathbf{S}^k$$

Graph VAR Models

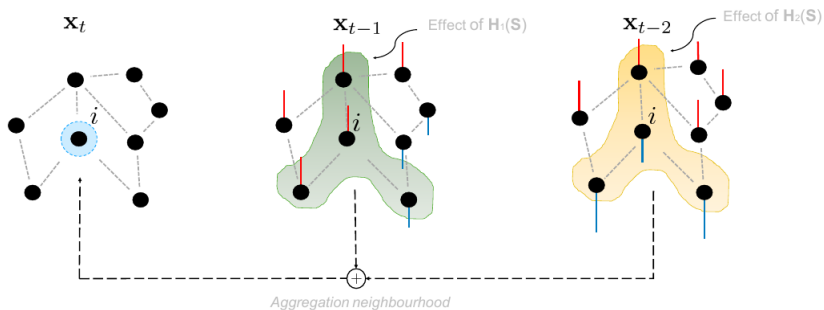
G-VAR model of order P: G-VAR(P)



- P graph filters: $P(K + 1)$ parameters

Graph VAR Illustration

- Time reach/order $P = 2$, graph reach/order $K = 1$



- Aggregate different time instant neighboring information

G-VAR Parameter Learning

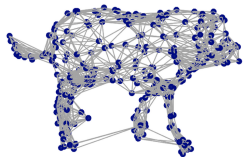
For forecasting

$$\min_{\mathcal{H}} \frac{1}{|\mathcal{T}| - P} \sum_{t=P+1}^{|\mathcal{T}|} [l(\mathbf{x}_t, \Phi(\mathbf{x}_t, \dots, \mathbf{x}_{t-P}, \mathcal{H}))] + \lambda g(\mathcal{H})$$

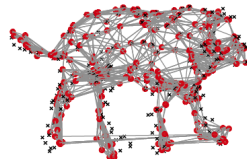
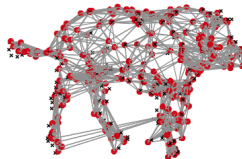
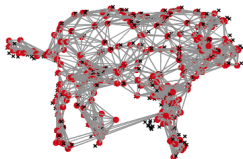
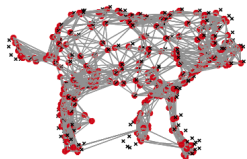
- **Model:** $\Phi(\mathbf{x}_t, \dots, \mathbf{x}_{t-P}, \mathcal{H}) = - \sum_{p=1}^P \sum_{k=0}^K h_{kp} \mathbf{S}^k \mathbf{x}_{t-p}$
- Parameters are filter coefficients $\mathcal{H} = \{h_{kp}\}_{kp}$
 - $\Rightarrow P(K+1)$
 - $\Rightarrow$ Transfer to any graph

Results

Walking dog dataset

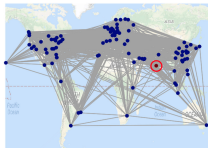


- Dataset: Walking dog mesh
 - ⇒ The mesh includes 251 points (nodes) over 59 time steps
 - ⇒ Graph is built by following 10-NN of coordinates at $t=1$
- **Task:** Predicting the average position along the three coordinates

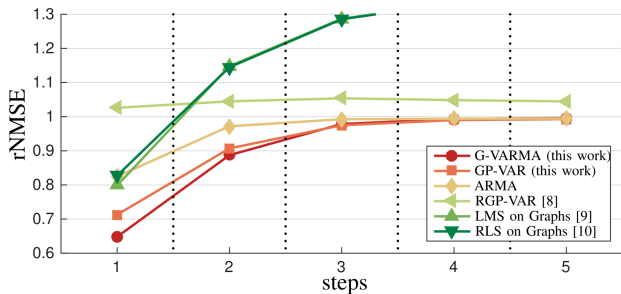


Machine Learning Applications

Epidemic diffusion dataset



- Dataset: Airline connectivity graph
 - ⇒ 100 nodes (airports) and 3,565 edges (flight connections).
 - ⇒ Initial state: infected node are in red; all others are susceptible (blue).
 - ⇒ Epidemic diffusion for $T = 122$ days.
- Task: Predict the daily evolution of the infection probability .



Topological Vector Autoregressive Models

Simplicial-Aware VAR models

Core idea. Start with the VAR

$$\mathbf{x}_t := \Phi_{\text{VAR}}(\mathbf{x}_{t-1}, \dots, \mathbf{x}_{t-P}; \mathcal{H}_{\text{VAR}}),$$

$$\mathcal{H}_{\text{VAR}} = \{\mathbf{W}_1, \dots, \mathbf{W}_P\},$$

substitute the parameter matrices with simplicial convolutional filters $\Rightarrow \mathbf{W}_p \leftarrow \mathbf{H}_p(\mathbf{L}_1)$

Simplicial-Aware VAR models

Core idea. Start with the VAR

$$\mathbf{x}_t := \Phi_{\text{VAR}}(\mathbf{x}_{t-1}, \dots, \mathbf{x}_{t-P}; \mathcal{H}_{\text{VAR}}),$$
$$\mathcal{H}_{\text{VAR}} = \{\mathbf{W}_1, \dots, \mathbf{W}_P\},$$

substitute the parameter matrices with simplicial convolutional filters $\Rightarrow \mathbf{W}_p \leftarrow \mathbf{H}_p(\mathbf{L}_1)$

► The Simplicial Vector Autoregressive model (S-VAR) is

$$\mathbf{f}_t = \sum_{p=1}^P \mathbf{H}_p(\mathbf{L}_1) \mathbf{f}_{t-p} + \varepsilon_t$$

$\Rightarrow$ Number of parameters drop from $\mathcal{O}(E^2)$ to $\mathcal{O}(L)$ $\Rightarrow$ from the number of edges to filter order

$\Rightarrow$ Inherits the topological properties of the simplicial convolutional filter

Simplicial-Aware VAR models

Core idea. Start with the VAR

$$\mathbf{x}_t := \Phi_{\text{VAR}}(\mathbf{x}_{t-1}, \dots, \mathbf{x}_{t-P}; \mathcal{H}_{\text{VAR}}),$$
$$\mathcal{H}_{\text{VAR}} = \{\mathbf{W}_1, \dots, \mathbf{W}_P\},$$

substitute the parameter matrices with simplicial convolutional filters $\Rightarrow \mathbf{W}_p \leftarrow \mathbf{H}_p(\mathbf{L}_1)$

- The Simplicial Vector Autoregressive model (S-VAR) is

$$\mathbf{f}_t = \sum_{p=1}^P \mathbf{H}_p(\mathbf{L}_1) \mathbf{f}_{t-p} + \varepsilon_t$$

$\Rightarrow$ Number of parameters drop from $\mathcal{O}(E^2)$ to $\mathcal{O}(L)$ $\Rightarrow$ from the **number of edges** to filter **order**

$\Rightarrow$ Inherits the **topological properties** of the simplicial convolutional filter

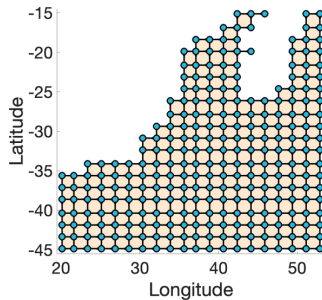
- Estimate the filter parameters by solving

$$\underset{\mathbf{h}}{\text{minimize}} \sum_{t=P+1}^{|\mathcal{T}|} [l(\mathbf{f}_t, \Phi_{\text{S-VAR}}(\mathbf{f}_t, \dots, \mathbf{f}_{t-P}, \mathbf{h}))] + \Omega(\mathbf{h}, \mathbf{L}_1)$$

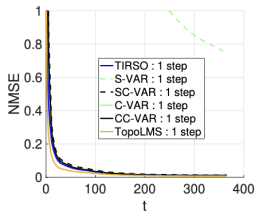
Applications

► Ocean currents network

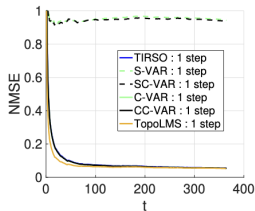
- ⇒ 225 nodes (0-cells), node signal: daily sea surface temperature
- ⇒ 498 edges (1-cells), edge signal: sea surface wind speed gradient
- ⇒ 274 polygons (2-cells), polygon signal: curl of the geostrophic current
- ⇒ Time-varying node, edge, and polygon signals for 2023



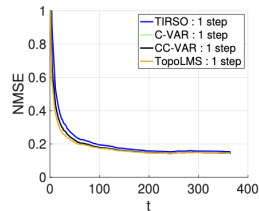
Ocean Currents Results



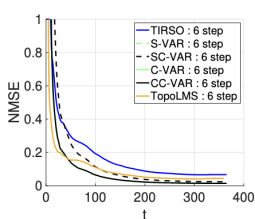
(a) 1-step ahead prediction of node signal (temperature)



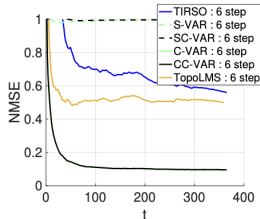
(b) 1-step ahead prediction of edge signal (wind speed gradient)



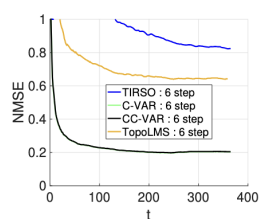
(c) 1-step ahead prediction of polygon signal (geostrophic current curl)



(e) 6-step ahead prediction of node signal (temperature)



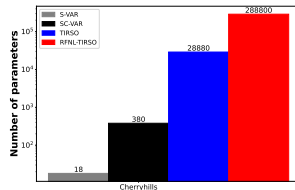
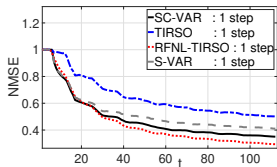
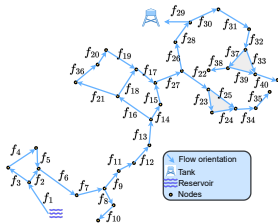
(f) 6-step ahead prediction of edge signal (wind speed gradient)



(g) 6-step ahead prediction of polygon signal (geostrophic current curl)

Applications

- ▶ Water distribution network
 - ⇒ 36 nodes (0-simplices), node signal: pressure
 - ⇒ 40 pipes (1-simplices), edge signal: flow rate
 - ⇒ **Time-varying** node signal pressure and edge flow signals



- ▶ Compares relatively well with topology-unaware alternatives
 - ⇒ while **reducing** parameter demand from $\sim 300K$ to **18**
 - ⇒ allows transference

Vertical coupling

- Real-world networks often have data defined over multiple simplicial levels

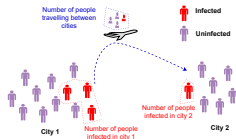


Figure. Epidemic spreading

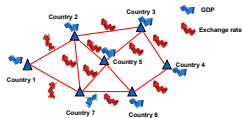


Figure. Financial Networks

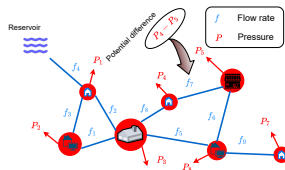


Figure. Water distribution

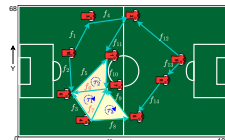


Figure. Multiagent systems

Joint presence of node and edge signals

- ▶ **Task** Model the joint evolution of node signal $\mathbf{v}_t$, edge flows $\mathbf{f}_t$, and triangle flows $\boldsymbol{\tau}_t$
- ▶ We could model node signals based on only node signals

$$\mathbf{v}_t = \sum_{p=1}^P \mathbf{H}_p^{00}(\mathbf{L}_0) \mathbf{v}_{t-p} + \boldsymbol{\epsilon}_{vt}$$

⇒ Similar to S-VAR, uses only signals of the same level

Joint presence of node and edge signals

- **Task** Model the joint evolution of node signal $\mathbf{v}_t$, edge flows $\mathbf{f}_t$, and triangle flows $\boldsymbol{\tau}_t$
- We could model node signals based on only node signals

$$\mathbf{v}_t = \sum_{p=1}^P \mathbf{H}_p^{00}(\mathbf{L}_0) \mathbf{v}_{t-p} + \boldsymbol{\varepsilon}_{vt}$$

⇒ Similar to S-VAR, uses only signals of the same level

- And add the influence of edge flows $\mathbf{f}_t$ by transforming with $\mathbf{B}_1$

$$\mathbf{v}_t = \sum_{p=1}^P \mathbf{H}_p^{00}(\mathbf{L}_0) \mathbf{v}_{t-p} + \mathbf{B}_1 \mathbf{H}_p^{01}(\mathbf{L}_1) \mathbf{f}_{t-p} + \boldsymbol{\varepsilon}_{vt}$$

Joint presence of node and edge signals

- **Task** Model the joint evolution of node signal $\mathbf{v}_t$, edge flows $\mathbf{f}_t$, and triangle flows $\boldsymbol{\tau}_t$
- We could model node signals based on only node signals

$$\mathbf{v}_t = \sum_{p=1}^P \mathbf{H}_p^{00}(\mathbf{L}_0) \mathbf{v}_{t-p} + \boldsymbol{\varepsilon}_{vt}$$

⇒ Similar to S-VAR, uses only signals of the same level

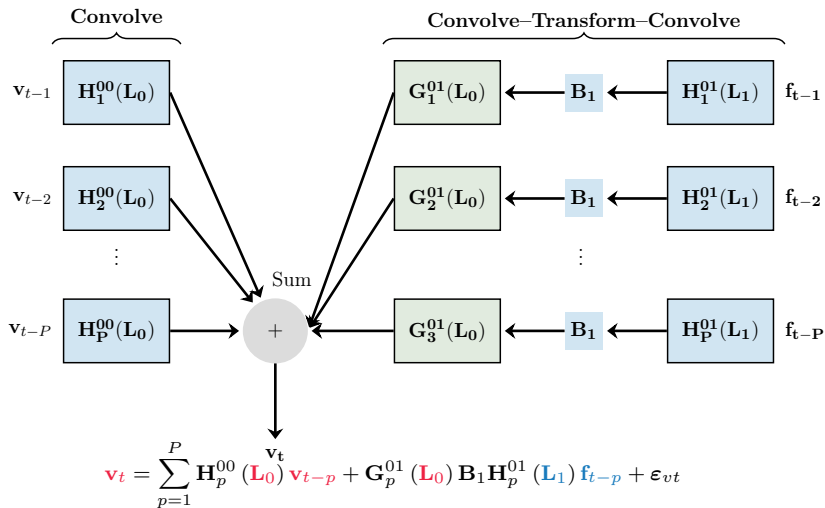
- And add the influence of edge flows $\mathbf{f}_t$ by transforming with $\mathbf{B}_1$

$$\mathbf{v}_t = \sum_{p=1}^P \mathbf{H}_p^{00}(\mathbf{L}_0) \mathbf{v}_{t-p} + \mathbf{B}_1 \mathbf{H}_p^{01}(\mathbf{L}_1) \mathbf{f}_{t-p} + \boldsymbol{\varepsilon}_{vt}$$

- And filter the transformed node signal (divergence) with a graph filter

$$\mathbf{v}_t = \sum_{p=1}^P \mathbf{H}_p^{00}(\mathbf{L}_0) \mathbf{v}_{t-p} + \mathbf{G}_p^{01}(\mathbf{L}_0) \mathbf{B}_1 \mathbf{H}_p^{01}(\mathbf{L}_1) \mathbf{f}_{t-p} + \boldsymbol{\varepsilon}_{vt}$$

Simplicial Complex-VAR (SC-VAR): node signals



SC-VAR: vertex, edge, triangle

- Node signals

$$\mathbf{v}_t = \sum_{p=1}^P \mathbf{H}_p^{00}(\mathbf{L}_0) \mathbf{v}_{t-p} + \mathbf{G}_p^{01}(\mathbf{L}_0) \mathbf{B}_1 \mathbf{H}_p^{01}(\mathbf{L}_1) \mathbf{f}_{t-p} + \boldsymbol{\varepsilon}_{vt}$$

SC-VAR: vertex, edge, triangle

- Node signals

$$\mathbf{v}_t = \sum_{p=1}^P \mathbf{H}_p^{00}(\mathbf{L}_0) \mathbf{v}_{t-p} + \mathbf{G}_p^{01}(\mathbf{L}_0) \mathbf{B}_1 \mathbf{H}_p^{01}(\mathbf{L}_1) \mathbf{f}_{t-p} + \boldsymbol{\varepsilon}_{vt}$$

- Edge signals

$$\mathbf{f}_t = \sum_{p=1}^P \mathbf{G}_p^{10}(\mathbf{L}_1) \mathbf{B}_1^\top \mathbf{H}_p^{10}(\mathbf{L}_0) \mathbf{v}_{t-p}$$

SC-VAR: vertex, edge, triangle

- Node signals

$$\mathbf{v}_t = \sum_{p=1}^P \mathbf{H}_p^{00}(\mathbf{L}_0) \mathbf{v}_{t-p} + \mathbf{G}_p^{01}(\mathbf{L}_0) \mathbf{B}_1 \mathbf{H}_p^{01}(\mathbf{L}_1) \mathbf{f}_{t-p} + \boldsymbol{\varepsilon}_{vt}$$

- Edge signals

$$\mathbf{f}_t = \sum_{p=1}^P \mathbf{G}_p^{10}(\mathbf{L}_1) \mathbf{B}_1^\top \mathbf{H}_p^{10}(\mathbf{L}_0) \mathbf{v}_{t-p} + \mathbf{H}_p^{11}(\mathbf{L}_1) \mathbf{f}_{t-p}$$

SC-VAR: vertex, edge, triangle

- Node signals

$$\mathbf{v}_t = \sum_{p=1}^P \mathbf{H}_p^{00}(\mathbf{L}_0) \mathbf{v}_{t-p} + \mathbf{G}_p^{01}(\mathbf{L}_0) \mathbf{B}_1 \mathbf{H}_p^{01}(\mathbf{L}_1) \mathbf{f}_{t-p} + \boldsymbol{\varepsilon}_{vt}$$

- Edge signals

$$\mathbf{f}_t = \sum_{p=1}^P \mathbf{G}_p^{10}(\mathbf{L}_1) \mathbf{B}_1^\top \mathbf{H}_p^{10}(\mathbf{L}_0) \mathbf{v}_{t-p} + \mathbf{H}_p^{11}(\mathbf{L}_1) \mathbf{f}_{t-p} + \mathbf{G}_p^{12}(\mathbf{L}_1) \mathbf{B}_2 \mathbf{H}_p^{12}(\mathbf{L}_2) \boldsymbol{\tau}_{t-p} + \boldsymbol{\varepsilon}_{ft}$$

SC-VAR: vertex, edge, triangle

- Node signals

$$\mathbf{v}_t = \sum_{p=1}^P \mathbf{H}_p^{00}(\mathbf{L}_0) \mathbf{v}_{t-p} + \mathbf{G}_p^{01}(\mathbf{L}_0) \mathbf{B}_1 \mathbf{H}_p^{01}(\mathbf{L}_1) \mathbf{f}_{t-p} + \boldsymbol{\varepsilon}_{vt}$$

- Edge signals

$$\mathbf{f}_t = \sum_{p=1}^P \mathbf{G}_p^{10}(\mathbf{L}_1) \mathbf{B}_1^\top \mathbf{H}_p^{10}(\mathbf{L}_0) \mathbf{v}_{t-p} + \mathbf{H}_p^{11}(\mathbf{L}_1) \mathbf{f}_{t-p} + \mathbf{G}_p^{12}(\mathbf{L}_1) \mathbf{B}_2 \mathbf{H}_p^{12}(\mathbf{L}_2) \boldsymbol{\tau}_{t-p} + \boldsymbol{\varepsilon}_{ft}$$

- Triangle signals

$$\boldsymbol{\tau}_t = \sum_{p=1}^P \mathbf{G}_p^{21}(\mathbf{L}_2) \mathbf{B}_2^\top \mathbf{H}_p^{21}(\mathbf{L}_1) \mathbf{f}_{t-p} + \mathbf{H}_p^{22}(\mathbf{L}_2) \boldsymbol{\tau}_{t-p} + \boldsymbol{\tau}_{ft}$$

SC-VAR: vertex, edge, triangle

- Node signals

$$\mathbf{v}_t = \sum_{p=1}^P \mathbf{H}_p^{00}(\mathbf{L}_0) \mathbf{v}_{t-p} + \mathbf{G}_p^{01}(\mathbf{L}_0) \mathbf{B}_1 \mathbf{H}_p^{01}(\mathbf{L}_1) \mathbf{f}_{t-p} + \boldsymbol{\varepsilon}_{vt}$$

- Edge signals

$$\mathbf{f}_t = \sum_{p=1}^P \mathbf{G}_p^{10}(\mathbf{L}_1) \mathbf{B}_1^\top \mathbf{H}_p^{10}(\mathbf{L}_0) \mathbf{v}_{t-p} + \mathbf{H}_p^{11}(\mathbf{L}_1) \mathbf{f}_{t-p} + \mathbf{G}_p^{12}(\mathbf{L}_1) \mathbf{B}_2 \mathbf{H}_p^{12}(\mathbf{L}_2) \boldsymbol{\tau}_{t-p} + \boldsymbol{\varepsilon}_{ft}$$

- Triangle signals

$$\boldsymbol{\tau}_t = \sum_{p=1}^P \mathbf{G}_p^{21}(\mathbf{L}_2) \mathbf{B}_2^\top \mathbf{H}_p^{21}(\mathbf{L}_1) \mathbf{f}_{t-p} + \mathbf{H}_p^{22}(\mathbf{L}_2) \boldsymbol{\tau}_{t-p} + \boldsymbol{\tau}_{ft}$$

- Compact form (SC-VAR)

⇒ Collecting in vector $\mathbf{x}_t = [\mathbf{v}_t^\top, \mathbf{f}_t^\top, \boldsymbol{\tau}_t^\top]^\top \in \mathbb{R}^{N_0+N_1+N_2}$ all signals

⇒ and in matrix $\mathbf{H}_p(\mathcal{L})$ all filters

$$\mathbf{x}_t = \sum_{p=1}^P \mathbf{H}_p(\mathcal{L}) \mathbf{x}_{t-p} + \boldsymbol{\varepsilon}_t$$

Theoretical results

Equivariances. The SC-VAR model is equivariant to both permutations and orientations.

Theoretical results

Equivariances. The SC-VAR model is equivariant to both permutations and orientations.

Topologically-aware. The SC-VAR model is topologically local.

- ▶ A simplicial process (e.g., $\mathbf{f}_t$) is **influenced** only by **vertically-adjacent** processes (e.g., $\mathbf{v}_t$ and τ_t)

Theoretical results

Equivariances. The SC-VAR model is equivariant to both permutations and orientations.

Topologically-aware. The SC-VAR model is topologically local.

- ▶ A simplicial process (e.g., $\mathbf{f}_t$) is **influenced** only by **vertically-adjacent** processes (e.g., $\mathbf{v}_t$ and τ_t)

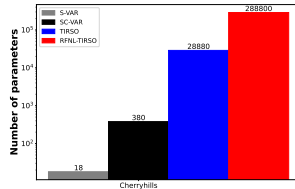
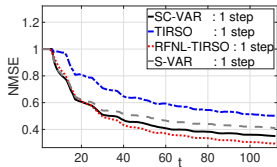
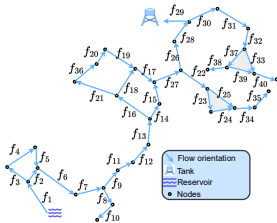
Joint spatiotemporal filtering. The SC-VAR

$$\mathbf{x}_t = \sum_{p=1}^P \mathbf{H}_p(\mathcal{L}) \mathbf{x}_{t-p} + \varepsilon_t$$

acts as a **two-dimensional** filter: **time** Fourier spectrum and **Hodge** Fourier spectrum.

Applications

- ▶ Water distribution network
 - ⇒ 36 nodes (0-simplices), node signal: pressure
 - ⇒ 40 pipes (1-simplices), edge signal: flow rate
 - ⇒ **Time-varying** node signal pressure and edge flow signals



- ▶ Joint processing **SC-VAR** improves on the simplicial-level processing (**S-VAR**)
 - ⇒ with an increase of parameter count 380

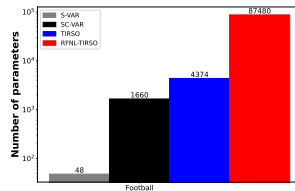
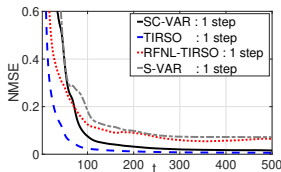
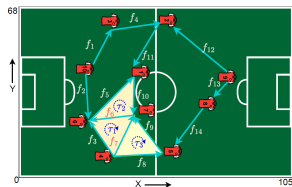
Applications

► Collaborative agents

⇒ 10 nodes (0-simplices), node signal: speed

⇒ 14 edges (1-simplices), edge signal: distance between agents

⇒ 3 triangle (2-simplices), triangle signal: area between agents



► Joint processing improves upon single level S-VAR

Learning Activity: Vertical Coupling in SC-VAR

Topological signals are often **vertically coupled**. For instance, in a water distribution network, node pressure is driven by the flow rates in incident pipes. The SC-VAR model uses **boundary operators** to bridge these levels.

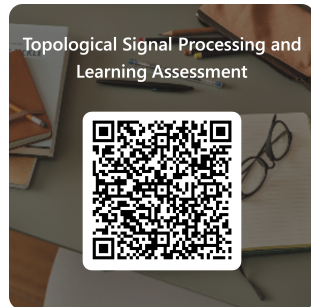
- **Q7.1:** You are building a forecasting model for a water network where you have historical data for both node pressures (v_t) and pipe flows (f_t). According to the SC-VAR equations, how is the influence of a past flow f_{t-1} transferred to the update of a current node signal v_t ?

- **Instructions**

- ⇒ Scan the QR code to open the activity
- ⇒ Work individually first
- ⇒ Then discuss your answer in pairs or groups of 3
- ⇒ Submit your final answer in Microsoft Forms

- **Logistics**

- ⇒ **Time:** 10 min
- ⇒ **Goal:** Understand how boundary operators enable information flow across different topological dimensions.
- ⇒ **Product:** One submitted response per student + justification to your peers



<https://forms.office.com/>

Further Reading

► General motivation references from where figures are taken

⇒ R. Bentivoglio, S. N. Jonkman, E. Isufi, and R. Taormina, “Probabilistic flood hazard mapping for dike-breach floods via graph neural networks,” *EGUsphere*, vol. 2025, pp. 1–29, 2025.

⇒ Z. Li, L. Ye, Y. Zhao, M. Pei, P. Lu, Y. Li, and B. Dai, “A spatiotemporal directed graph convolution network for ultra-short-term wind power prediction,” *IEEE Transactions on Sustainable Energy*, vol. 14, no. 1, pp. 39–54, 2022.

⇒ C. Zheng, X. Fan, C. Wang, and J. Qi, “Gman: A graph multi-attention network for traffic prediction,” in *Proceedings of the AAAI conference on artificial intelligence*, vol. 34, no. 01, 2020, pp. 1234–1241.

⇒ Z. He, P. Chen, X. Li, Y. Wang, G. Yu, C. Chen, X. Li, and Z. Zheng, “A spatiotemporal deep learning approach for unsupervised anomaly detection in cloud systems,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 34, no. 4, pp. 1705–1719, 2020

⇒ H. Hojjati, M. Sadeghi, and N. Armanfard, “Multivariate time-series anomaly detection with temporal self-supervision and graphs: Application to vehicle failure prediction,” in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*. Springer, 2023, pp. 242–259

⇒ H. Liu, D. Yang, X. Liu, X. Chen, Z. Liang, H. Wang, Y. Cui, and J. Gu, “Todynnet: temporal dynamic graph neural network for multivariate time series classification,” *Information Sciences*, vol. 677, p. 120914, 2024

Further Reading

► Topological complexes

⇒ A. R. Benson, R. Abebe, M. T. Schaub, A. Jadbabaie, and J. Kleinberg, “Simplicial closure and higher-order link prediction,” *Proceedings of the National Academy of Sciences*, vol. 115, no. 48, 2018

⇒ S. Barbarossa and S. Sardellitti, “Topological signal processing over simplicial complexes,” *IEEE Transactions on Signal Processing*, vol. 68, pp. 2992–3007, 2020

► Graph and topological VAR

⇒ Isufi, E., Loukas, A., Perraudin, N., & Leus, G. (2019). Forecasting time series with VARMA recursions on graphs. *IEEE Transactions on Signal Processing*, 67(18), 4870-4885.

⇒ Krishnan, J., Money, R., Beferull-Lozano, B., & Isufi, E. (2024). Simplicial vector autoregressive models. *IEEE Transactions on Signal Processing*, 72, 5454-5469.

⇒ Canbolat, A., Money, R., & Beferull-Lozano, B. (2025, September). Cellular Autoregressive Higher-Order Models. In *2025 33rd European Signal Processing Conference (EUSIPCO)* (pp. 2517-2521). IEEE.